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Application of internet of things (IoT) and machine learning (ML) in intelligent logistics supply chain

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Abstract

Supply chain (SC) logistics efficiency is vital for competitiveness in today's data-driven global economy. This study presents an intelligent, predictive logistics framework integrating the Internet of Things (IoT) and Machine Learning (ML) to enhance real-time decision-making. Conventional SC management approaches struggle to address the complexity and dynamism of modern logistics systems; hence, this research introduces the Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model. The framework employs IoT-sensed data for continuous learning, enabling adaptive demand forecasting that directly supports procurement, inventory management, and transportation planning. Using the publicly available “Supply Chain Dataset of a Multi-Product Retailer” from Kaggle, the data were preprocessed and modeled through the LHHBO algorithm for hyperparameter optimization within a Deep Neural Network. Experimental results, implemented in Python, demonstrate that the proposed LHHBO-DNN model achieves higher forecasting accuracy and robustness compared with conventional predictive methods. The findings highlight the potential of IoT–ML integration to improve supply chain responsiveness, minimize downtime, and enhance overall logistics efficiency.

Article highlights

- The study demonstrates the strong potential of integrating Internet of Things (IoT) technologies with machine learning (ML) to enhance supply chain responsiveness, reduce operational downtime, and improve overall logistics efficiency.
- Supply chain management (SCM) is emphasized as a critical function for coordinating interdependent organizations to ensure the smooth flow of goods, information, and capital.
- The paper illustrates the practical implementation of real-time IoT data streams combined with ML-based predictive analytics to enable intelligent and data-driven logistics management.

Keywords Logistics, Supply chain, Demand forecasting, Machine learning, Performance, IoT



1 Introduction

The Internet of Things (IoT) has transformed logistics operations by enabling real-time sensing, monitoring, and data exchange across the supply chain (SC). Foundational studies such as Al-Fuqaha et al. [34] provided a comprehensive survey of IoT architecture, communication protocols, and key application domains, offering a theoretical basis for understanding how IoT technologies support data-driven logistics and supply-chain management. Early implementations, such as Radio Frequency Identification (RFID) systems, evolved from manual data collection toward intelligent, automated tracking networks. Integrating IoT with warehouse and transportation systems improves visibility, accuracy, and decision-making by continuously monitoring vehicle speed, fuel consumption, temperature, and inventory levels [1–3]. This transition enhances storage efficiency, reduces labor costs, and optimizes goods distribution and delivery. Artificial Intelligence (AI) and Machine Learning (ML) further extend these capabilities by processing large volumes of IoT data to uncover operational patterns and optimize routing, demand forecasting, and resource allocation [4, 5]. Collectively, these technologies have redefined modern logistics and warehousing systems as interconnected, data-driven ecosystems that enhance both efficiency and sustainability.

Supply chain management (SCM) plays a vital role in coordinating interdependent organizations to ensure the seamless flow of goods, information, and capital. However, the growing globalization and volatility of market demand have increased complexity and risk within supply chains [6–9]. Traditional SCM approaches, often based on static forecasting and manual decision processes, lack adaptability to dynamic environments. Information Technology (IT) acts as a critical enabler by improving data acquisition, communication, and decision support across the supply chain [10]. Within this context, IoT has emerged as a core driver of Industry 4.0, creating interconnected networks that enhance supply chain transparency, agility, and resilience [11, 12]. Despite these advances, existing IoT-enabled models often lack adaptive predictive mechanisms that can learn from real-time data streams, limiting their ability to respond to sudden market changes.

This study focuses on demand forecasting as a central challenge in supply chain logistics, where prediction accuracy directly influences procurement, inventory management, and transportation planning. To address the limitations of conventional predictive systems, we propose a novel Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) framework. The model integrates IoT-sensed data with a bio-inspired optimization algorithm that autonomously tunes the parameters of a Deep Neural Network (DNN). This hybrid architecture enables continuous learning from real-time logistics data, improving forecasting accuracy and operational responsiveness under fluctuating supply and demand conditions.

The primary contributions of this study are fourfold. First, it introduces the LHHBO-DNN model, which combines IoT-enabled data acquisition with adaptive machine learning for intelligent logistics management. Second, it demonstrates how swarm-intelligence optimization enhances model generalization, convergence speed, and stability in dynamic supply chain environments. Third, by employing a publicly available dataset from Kaggle and benchmarking against ensemble and deep learning baselines, the research establishes a reproducible framework for comparative evaluation. Finally, the study highlights practical implications for enhancing demand forecasting, inventory

control, and transportation planning, thereby supporting data-driven decision-making and supply chain resilience. Overall, the proposed hybrid IoT–ML approach contributes to the development of intelligent, autonomous, and adaptive supply chain systems aligned with the principles of Industry 4.0.

The motivation for this study stems from the persistent challenges of maintaining forecasting accuracy and operational efficiency under rapidly changing supply and demand conditions. Factors such as shorter product life cycles, shifting consumer behavior, and global disruptions have intensified the need for predictive systems capable of learning from real-time IoT data. Traditional forecasting approaches, which rely heavily on static historical information, often lead to overstocking, shortages, and inefficiencies in logistics planning. From a practical standpoint, organizations now require intelligent forecasting tools that can dynamically adjust to evolving market signals, thereby improving inventory management, cost efficiency, and service responsiveness.

From a research perspective, although numerous studies have explored the use of Machine Learning (ML) and the Internet of Things (IoT) in supply chain analytics, existing frameworks are typically limited to isolated predictive models or static optimization strategies. There remains a critical gap in developing unified, adaptive models that can integrate real-time IoT data streams with intelligent learning mechanisms for continuous forecasting refinement. This study is motivated by the need to bridge this gap by proposing the Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO–DNN) model, which combines bio-inspired optimization and deep neural learning for dynamic, data-driven forecasting in intelligent logistics supply chains.

This paper shows how real-time Internet of Things (IoT) data streams can be practically implemented together with machine-learning-based predictive analytics to manage intelligent logistics. Through a mixture of constant IoT sensing, coupled with a refined deep neural network, the suggested structure allows to take into account the demand and make a decision in procurement, inventory management, and transportation planning. The study has made empirical findings that IoT-based predictive models are more efficient in terms of responsiveness, precision of forecasts and efficiency of operations than traditional supply chain solutions.

The primary objectives of this research are threefold:

- (i) to develop a hybrid IoT–ML framework, termed Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO–DNN), that enhances predictive accuracy and adaptability in supply chain demand forecasting;
- (ii) to evaluate the performance of the proposed model against conventional and ensemble learning approaches using real-world logistics data; and
- (iii) to demonstrate how the integration of IoT-sensed data with bio-inspired optimization can support intelligent decision-making in demand forecasting, inventory control, and transportation management. These objectives collectively aim to advance both theoretical understanding and practical implementation of intelligent, data-driven logistics systems.

2 Literature review

Supply chain management (SCM) ensures that the correct products and quantities reach the right locations at the right time and cost [13]. However, persistent challenges such as delivery delays, overstocking, and stockouts continue to hinder operational efficiency.

The integration of analytics, business intelligence, and smart technologies has therefore become essential to enhance supply-chain resilience and value creation [14]. In an increasingly globalized and competitive environment, organizations are adopting fog, edge, and cloud computing in conjunction with big data analytics and the Internet of Things (IoT) to improve coordination, reduce disruptions, and enable seamless information exchange across their networks [15, 16].

Extensive research has explored the application of IoT and artificial intelligence (AI) in logistics and warehousing, including intelligent distribution systems, warehouse automation, and real-time tracking [17–22]. Technologies such as Wireless Sensor Networks (WSN), blockchain, and AI-based analytics have improved logistics visibility, traceability, and decision-making [18–21]. Despite these advances, challenges remain regarding device cost, interoperability, and data security [22, 23]. To address these limitations, recent studies have combined IoT data streams with machine learning (ML) techniques to enhance logistics intelligence and predictive capabilities.

Machine learning and AI-driven analytics have shown strong potential in forecasting and decision support within supply chains. Rahman et al. [29] and Feizabadi [30] demonstrated that ML algorithms substantially improve demand forecasting accuracy and overall supply-chain performance. Complementary research by Cakiroglu et al. [31] applied interpretable ensemble learning with SHAP analysis to improve explainability and reliability in predictive modeling—an approach relevant to transparent supply-chain analytics. Aylak [32] proposed the SustAI-SCM framework integrating agentic AI for sustainable and cost-efficient logistics automation, while Aylak et al. [33] and Aylak and Oral [34] analyzed the digital transformation of logistics enterprises, highlighting the growing adoption of AI-based solutions for operational efficiency.

Further, several studies have underscored the role of predictive analytics in managing uncertainty and demand fluctuations within supply chains. Notable examples include research on forecasting analytics for e-fulfilment error assessment, pandemic-related disruptions, big data applications in demand forecasting, and reviews of ML techniques for predictive supply-chain management. Collectively, these works demonstrate the growing importance of integrating IoT and ML to improve responsiveness and forecasting precision. However, most existing approaches focus on isolated predictive tasks or single-domain optimizations rather than holistic, adaptive frameworks.

Building on these insights, the present study proposes a Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model that unifies IoT-driven data acquisition with bio-inspired optimization and deep neural learning. This adaptive hybrid framework addresses the identified research gap by enabling continuous, data-driven forecasting and real-time decision support across logistics and supply-chain operations.

In summary, existing research on IoT- and AI-driven logistics primarily focuses on isolated predictive tasks or single-domain optimization rather than fully integrated, adaptive systems. Most current frameworks lack the ability to learn continuously from real-time IoT data streams and dynamically recalibrate forecasting models under changing market conditions. To address this critical gap, the present study develops the Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model, which unifies IoT-enabled data acquisition with bio-inspired optimization and deep learning to enable adaptive, data-driven forecasting across supply-chain operations.

Table 1 Evaluations of various parameters

Parameters	Equations
Recall	$Precision = \frac{TP}{FP+FN}$
F1-score	$F1 - Score = 2 * \frac{(Precision * Recall)}{(Precision + Recall)}$
AUC	$AUC = \int_0^1 TP(FP) d(FP)$
Precision	$Precision = \frac{TP}{FP+TP}$
Accuracy	$Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)}$

True negative (TN) and true positive (TP) can accurately predict the cognitive neuroscience classification. The result of a false negative (FN) for cognitive neuroscience and an inaccurate diagnosis of cognitive neuroscience is a false positive (FP)

3 Materials and methods

This section outlines the methodological framework adopted to design, develop, and evaluate the proposed Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model for intelligent supply-chain forecasting. The research process follows a structured sequence comprising five stages: (i) data collection from a publicly available Kaggle dataset; (ii) data cleaning and preprocessing to ensure accuracy and consistency; (iii) feature extraction and transformation using statistical and image-based methods; (iv) model development, where the Honey Bees Optimization algorithm is integrated with a Deep Neural Network for parameter tuning and learning; and (v) model evaluation and comparative analysis against existing ensemble and deep learning approaches. This methodological flow ensures a systematic assessment of the proposed framework's effectiveness in enhancing demand forecasting and logistics decision-making. The evaluation of the parameters is shown in Table 1.

The collected supply chain (SC) dataset from Kaggle was then subjected to cleansing and min–max normalization to ensure consistency and reliability. To improve SC management and forecast future demand levels, the study employed a logistics-driven Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model. A multi-model strategy was also utilized, incorporating ARIMAX–neural networks, Artificial IoT, Conditional Generative Adversarial Networks (CGAN), Generative AI, and Voting Regressors. The objective was to integrate the strengths of these models to enhance decision-making accuracy across diverse product portfolios. This hybrid approach bridges critical knowledge gaps and supports the argument that complex supply-chain challenges require advanced, adaptive machine-learning techniques. The performance analysis of different methods is shown in Table 2.

3.1 Dataset

We used the Kaggle platform to collect data on the supply chain. Orders, goods, inventory, suppliers, logistics, and demand are all part of the SC process data set. To optimize

Table 2 Performance analysis of different methods

Methods	Parameters				
	F1-Score (%)	Precision (%)	AUC (%)	Recall (%)	Accuracy (%)
AIoT	85.5	82.6	87.5	83.6	89.9
GAI	74.4	73.5	77.2	72.3	73.2
CGAN	95.0	94.5	94.70	92.5	94.7
Voting Regressor	73.4	83.6	76.5	71.4	72.6
ARIMAX-NN	79.1	80.1	79.4	78.5	78.4
Proposed DNN	98.3	95.3	98.3	94.4	96.5

SC effectiveness and performance, it employs analytical forecasting, inventory control, and logistics optimization.

The dataset employed in this research was sourced from the publicly available Kaggle repository titled “Supply Chain Dataset of a Multi-Product Retailer” (<https://www.kaggle.com/datasets/>). It contains approximately 65,000 transactional records covering product categories, order quantities, inventory levels, lead times, delivery schedules, and supplier details over multiple fiscal periods. Each record includes 14 key attributes such as product ID, date, location, demand volume, stock quantity, and transit duration. This dataset was selected because it provides a comprehensive representation of end-to-end supply chain operations, enabling detailed modeling of demand forecasting and logistics optimization. Its scale, diversity, and inclusion of time-dependent demand information make it suitable for evaluating deep learning-based predictive models such as the proposed LHHBO-DNN framework.

3.2 Data cleaning

Data cleaning on the SC dataset requires multiple steps to ensure accurate and trustworthy results. We first considered using methods of imputation to fill in the gaps caused by missing data or removing the affected rows entirely. Find any discrepancies in the order format, supplier information, or names, and fix them. Get rid of duplicates so your analysis isn't biased. The accuracy of numerical data types and the standardization of date formats are both important for maintaining consistency. Lastly, to guarantee the dataset's general quality, we cross-check it with pre-existing benchmarks or internal criteria to make sure the data is accurate.

3.3 Pre-processing using histogram equalization

The local elements can be obtained via edge operators by computing regional data like the local standard deviation, mean, etc. Local approaches use a feature-based methodology. The study provides contrast amplification by changing the characteristics. The typical pre-process-based approach starts by defining the contrast before boosting the contrast ratio to improve image contrast. A different approach uses local histogram alteration to improve local image contrast.

The key concept is to establish each pixel's immediate action (W_0, Z_0) depending on its neighbours in a constrained window centred at this pixel:

$$j_0(W_0 - Z_0) = e(J(w, z)) \quad (1)$$

They are quite good in improving regional texture. However, as the result is not a monotone mapping because the original image's order of grey levels may be significantly altered, they can blur the original pictures. The majority of global methods make use of histogram modification techniques.

Histogram equalisation (HE) is one of the most widely used techniques. The basic goal of HE-based approaches is to assign new intensities to pixels in order to maximally uniformize the brightness pattern. Assume that the original image has been normalised, that its range of intensities is, that $o(w)$ is a density function of the original image's intensity distribution, and that (w) stands for the photograph's normalized value for intensity. After equalisation, the output image's desired density function of intensity distribution equals 1, or the HE equation is

$$Z = \int_0^w o(w) du \quad (2)$$

where u is an imaginary variable, $o(w)$ is the contrast of an input reputation, and $o(w)$ is the strength of the resulting image. $[0, 1]$ includes w and 0 .

When an image comprises a single item or when there isn't a noticeable contrast difference between the thing in question and the backdrop can HE be used to enhance low contrast images. Since the aforementioned requirements cannot always be satisfied, the global approaches have issues with over- and under-enhancement.

Numerous modifications have been created to address the aforementioned shortcomings. Assume that the range of grey levels is $[x_0, x_L]$, and that two or many mid-nodes x_i ($i = 1, 2, \dots, L-1$) were calculated using the image histogram's median, mean, and maximum or amount of spikes. The original histogram is divided into segments between $[x_i, x_{i+1}]$, and all of them will be individually and piecewise equalised. This technique, known as multi-peak HE, can just slightly improve HE approach and still retains its flaws.

3.4 Feature extraction using scale-invariant feature transform

A well-known technique that effectively integrates both ideas is the Scale Invariant Feature Transform (SIFT), which Lowe first proposed. It takes the extremes of the Difference-of-Gaussians into account for interest points and a histogram of directions for local descriptors. The SIFT algorithm discovers extreme in the response time of the image's integration with a DoG function by detecting points using a scale-invariant manner.

$$\psi(y, x, \sigma) = s(y, x, \sigma) - s(y, x, \sigma) \quad (3)$$

where the two-dimensional Poisson kernel $F(y, x)$ has an accepted deviation of. It is possible to compute the convolution of a photograph with (y, x) as the difference of successive images in the scale-space that represents the image by appropriately selecting a number of k . Two distinct definitions of the scale-space description $L(y, x, d): \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}$ of an image $I(y, x)$ are equal. The first one illustrates how the heat distribution $I(y, x)$ has changed over the years in an infinitely uniform medium:

$$\partial_d F(y, x, d) = \frac{1}{2} \nabla^2 F(y, x, d) \quad (4)$$

$s(y, x, \sigma) = J(y, x)$ is the first circumstance in this scenario. The next step is the image's subsequent integration with a Poisson kernel, which is also $g(x, y)$, whose standard error is t :

$$F(y, x, \sigma) = s(y, x, \sigma) * J(y, x) \quad (5)$$

Due to the separateness of the Gaussian filter, this scale-space picture of an image is effectively generated using the definition of the convolution. Then, the descriptors for SIFT are computed using the nearby data for each interest point. These local characteristics are not affected by scale or rotate changes. They are made up of a three-dimensional in nature histogram with two spatial and one oriented dimension. The scale at which the point was discovered determines the size in this region. It has been used to

handle situations requiring track or matching of regions because of the tool's simplicity, strong repeatability, and precision on matching findings.

3.5 Logistics based hybrid honey bees optimized improved convolutional neural network (HHBO-ICNN)

The proposed framework, referred to as the Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN), integrates a bio-inspired optimization mechanism with a deep learning architecture. In this model, the Hybrid Honey Bees Optimization (HHBO) algorithm fine-tunes the neural network's parameters to enhance prediction accuracy and convergence efficiency. Although earlier conceptual iterations explored convolutional extensions, the finalized architecture employs a Deep Neural Network (DNN) due to its superior adaptability for multivariate, time-dependent forecasting tasks in logistics. The terminology “LHHBO-DNN” is therefore used consistently throughout this study to denote the complete hybrid model.

3.5.1 Logistics based hybrid honey bees optimized

Hybrid Honey bees is a IoT algorithm which can be inspired by natural phenomena like ant colonies and bee swarms, these algorithms are used for decentralized and self-organizing solutions in complex IoT environments. Hybrid honey bees optimized that simulate the foraging behave of a honey bees optimized. Three sorts of bees employed honeybees, spectator bees, and scout bees—are tasked with feeding in the hive. The employees are responsible for transporting nectar recently exposed supplies to the swarm and spinning inside the hive to inform intrigued bees sitting in the bee about those profitable sites. The watching bees track the hired bees' activities and decide on a source to take benefit of. Scout bees look for untapped reserves using either internal incentive or an outside hint. In simple terms, employed and observer beekeepers are in charge of taking use of food reports, whereas scout bees are in charge of discovering fresh sources of food. Each source of nourishment refers to a solution in a D-Optimization problem with dimensions in the optimization point of view, and source's nectar content represents the solution's health value. Each utilized bee explores in the vicinity of its food source from memory throughout the exploration-exploitation process, and each onlooker bee searches in the vicinity of the food source based on the knowledge supplied by using bees and waggle dances. The following are the HHBO's fundamental steps. The following are the fundamental ABC steps. Food sources are initialized by Eq. (6):

$$W_{ji} = W_i^{min} + rand(0,1) (W_i^{max} - W_i^{min}) \quad (6)$$

where j is a feeding source's location in a spectrum of 1 and D , where D is the size of searching area, and is a meal source's value in the area of 1 and SN , where SN is the population size; W_i^{min} and W_i^{max} is Positional upper and lower boundaries j . Each bee in use develops its own food supply by Eq. (7):

$$U_{ji} = W_{ji} + \varnothing_{ji}(W_{ji} - W_{li}) \quad (7)$$

where W_j the existing food supply; W_l is the chosen supply of food for W_{ji} ; j is those in position to disturb at random; u_j food supply that has evolved as a result of j th variable

evolution w_j ; and $\varnothing_{ji}$ is an evenly distributed number between -1 and 1 . Apply greedy selection between

$$u_j \text{ and } W_j. \text{ if } e(u_j) > e(W_j) \quad (8)$$

The worker bee departs w_j and memorizes u_j as the current source. A coefficient of likelihood has been given to each dietary source by Eq. (9).

$$o_j = \frac{fitness_j}{\sum_{j=1}^{SN} fitness_j} \quad (9)$$

In which case is source's efficiency level W_j and Size of the populace is SN. Each observer bee makes a probabilistic choice for an energy supply after which it searches as in the worker bee step. The scouting bee creates another source of food using Eq. (6) if there is any depleted food source that is defined by a "Limit" value. Instead of abandoned one. To reach the optimum set of cycles, repeat steps 2 through 6 as necessary (Fig. 1).

3.5.2 Deep neural network- DNN

Latest Artificial Intelligence (AI) applications that are formed of numerous hidden layers and neural nodes are based on DL. It is widely used in neural networks for image

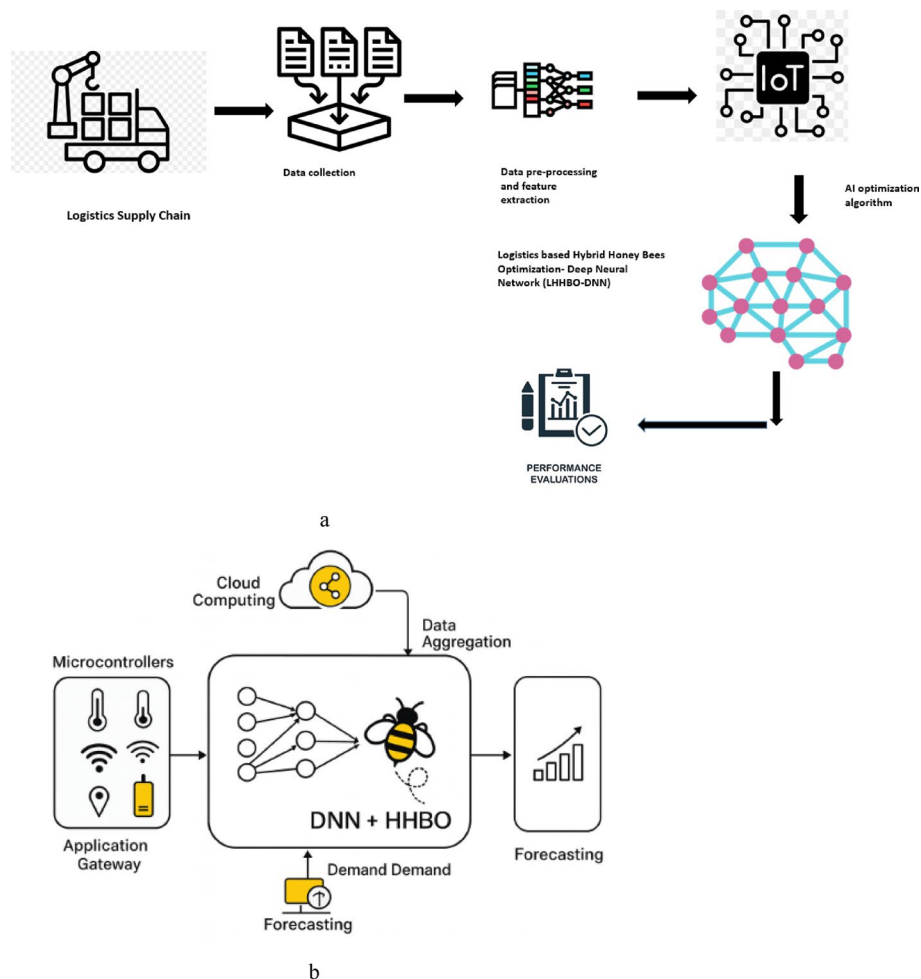


Fig. 1 **a** Proposed model. **b** The proposed framework and data flow from IoT sensors to predictive outputs

identification, and other fields and certain improvements in communication have been established. With the rapid development of DL, the conventional approach of modulation identification of digital signals possesses an increasingly given method to the proposed DNN method. The multidimensional information vector that is shown on the visible layer serves as the input for the proposed DNN, also known as the training illustration. Every hidden layer performs a sequence of non-linear changes and is characterized by the following Eq. (10):

$$B = \text{sig}(Z * X + y) \quad (10)$$

where Z and y represent the encoding weight matrix and bias vector, respectively, and sig stands for sigmoid activation function. Each neural node's input is represented by X . The output error is provided in Eq. (11), calculated using the mean square error function after having the stochastic gradient descent approach for training the hidden layers.

$$f_{\text{output}} = \frac{1}{m} \sum_{j=1}^m \sum_{i=1}^I (X'_{ij} - X_{ij})^2 + \lambda * \Omega_{\text{weight}} \quad (11)$$

The amount of weight Ω_{weight} is applied to the cost function's weights and can be expressed as follows Eq. (12): where X'_{ij} is the actual output following a sequence of operations in demand forecasting, λ maps the coefficients for the K_2 regularization.

$$\Omega_{\text{weight}} = \frac{1}{2} \sum_l^K \sum_i^W \sum_j^N (Z_{ij}^l)^2 \quad (12)$$

where K and N stand for the amount of input variables and hidden layers. The test data can be provided into the proposed DNN model after it has been trained to develop predictions and transform the output to recognition accuracy. Figure 2 illustrates the proposed DNN algorithm. The proposed Deep Neural Network (DNN) is a fully connected feed-forward model that is to be used in demand and logistics forecasting based on IoT-enabled supply chain data. It includes an input layer, which takes normalized numerical features and encoded categorical variables and four hidden layers, which have 128, 64, 32 and 16 neurons, respectively. Each of the hidden layers is trained with the ReLU activation function to effectively learn non-linear interactions between variables of the supply chain, and dropout (0.2) and L2 regularization are used to avoid overfitting. The single neuron of the output layer has linear activation function, which is appropriate in the continuum value prediction. The network is optimized through Adam optimizer and loss function of mean squared error (MSE) and hyperparameters optimally adjusted through Hybrid Honey Bees Optimization (HHBO) algorithm to enhance accuracy and reliability of forecast.

4 Results and discussion

We examined a range of interference frequencies. Our implementation employed a publicly accessible framework based on Python and was trained for eighteen hours on an NVIDIA K40 GPU computer. The proposed N classifies cognitive neuroscience by demonstrating many measurement metrics in a comparable approach, including an area under the curve (AUC), f1- score, accuracy, precision and recall score, as shown in Table 3. For validating the proposed model approaches efficacy, the comparison of existing

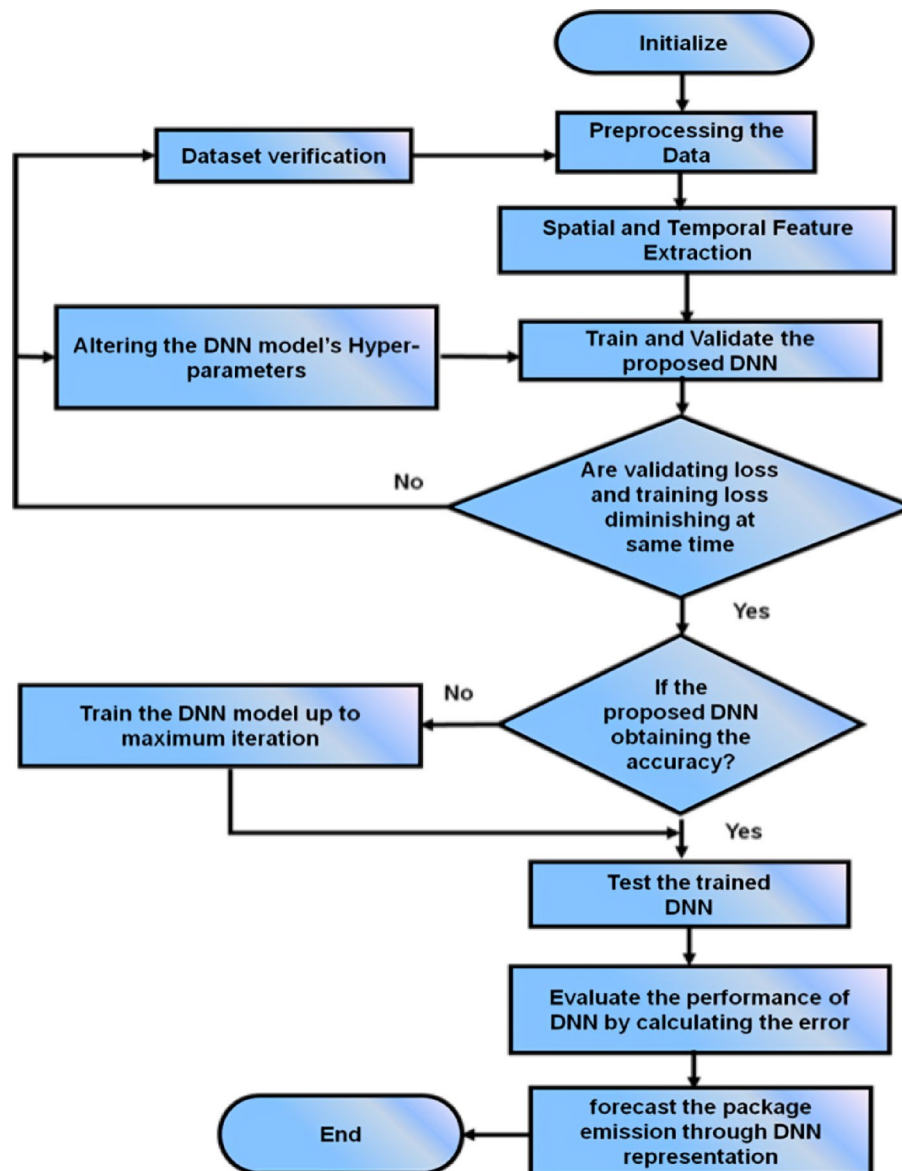


Fig. 2 Flow DNN Algorithm

methods such as AIoT [21], GAI [22], CGAN [23], Voting Regressor [24], ARIMAX-DNN [25] were demonstrated.

The model has been utilized in the sequence of validation for evaluating the performance of trained model after training the whole training data set.

Precision is frequently utilized as an outcome variable, particularly with classification tasks. A model that determines the quantity of real positive predictions also known as reliable prediction based on the overall number of optimistic forecasts generated by the instance. The proposed provided a precision of 95.3%. AIoT has 82.6%, GAI (73.5%), CGAN(94.55), Voting regressor (83.6) and ARIMAX-NN demonstrates 80.1% % of precision. Compared to the [26] accuracy and precision of various techniques, the proposed model determines the improved outcomes than the existing methods. The proposed approach performs more efficiently than the existing methods (Fig. 3).

Table 3 References

References	Objectives	Methods	Limitation
[16]	Among the many uses of the Internet of Things (IoT) and artificial intelligence (AI) in contemporary logistics and warehousing are the following: intelligent distribution system optimization, intelligent warehouse management system construction, intelligent real-time tracking and monitoring of logistics information, and comprehensive AI applications in supply chain management.	Block chain	The study has limited dataset for exploring the application of AI in supply chain management.
[17]	Examining the current supply chain system and the advantages gained by implanting smart items and embedded networks of the Internet of Things is the goal of this article.	Wireless Sensor Networks (WSN)	Smart devices, particularly RFID, are prohibitively costly, which prevents their widespread use.
[18]	Focusing on automated warehousing and unmanned delivery technologies, this work investigates how intelligent logistics systems might be used to supply chain management.	Machine Learning	One of the most important concerns when implementing intelligent logistics systems is data security.
[19]	This article explores the process of creating a blockchain-based intelligent logistics analysis platform.	Blockchain	With the help of Internet of Things (IoT) technology, intelligent logistics can be accomplished, but only after the data has traversed various administrative borders and served various, possibly unanticipated, objectives.
[20]	Automated warehousing and unmanned delivery systems are the focal points of this investigation into the use of intelligent logistics systems in SCM.	Prediction Model	Inadequate authorization, unencrypted transport, unsecured Web interface, and privacy concerns are the most prevalent and easily fixable security issues.
[21]	Path planning optimization, intelligent warehousing sorting, and real-time monitoring and early warning are some of the important AI applications discussed in this article, which focuses on smart logistics scheduling.	Path Optimization	It communicates and exchanges a great deal of personal information, product information, and corporate information via the Internet as part of the logistical process.
[22]	Using the Internet of Things (IoT) as an example, we present a literature review on smart logistics.	IoT	The Internet of Things (IoT) presents both opportunities and threats for the future of smart logistics but the study chose limited references.
[23]	In the context of edge computing (EC) and the Internet of Things (IoT), this research seeks to optimize the architecture of intelligent logistics systems and supply chain management.	Edge computing-IoT	A wide range of protocols, data formats, and physical sensing resources must be managed throughout the operation process to accommodate the expansion of IoT technologies in smart logistics.

Recall that can be referred to as sensitivity or the average of actually appropriate instances in the database, quantifies the rate at which the model detects every potential case. Divide the whole amount of real positives by the overall amount of positive instances to arrive at the total. The F-measure is the mean recall and precision after considering [27] false negatives and inaccurate outcomes. It performs effectively when the

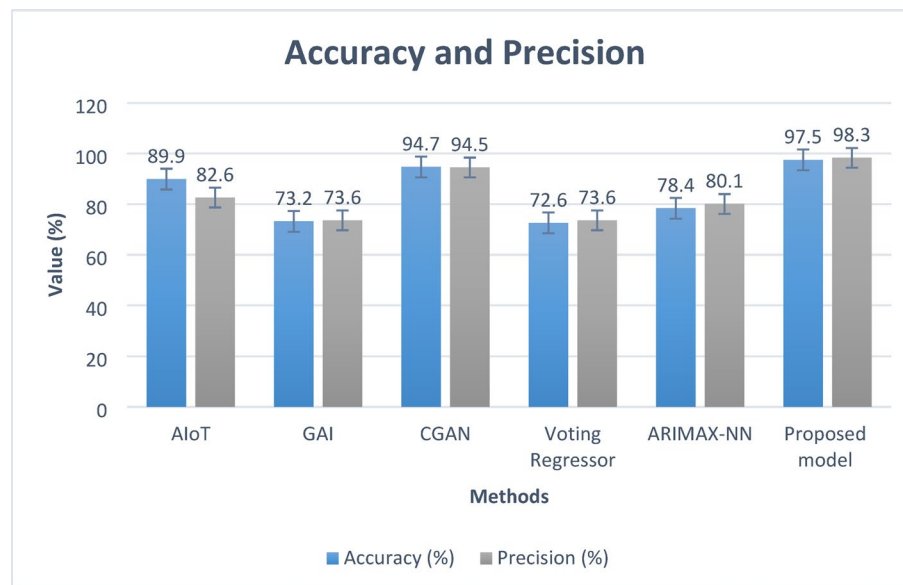


Fig. 3 Comparison Outcomes of accuracy and precision

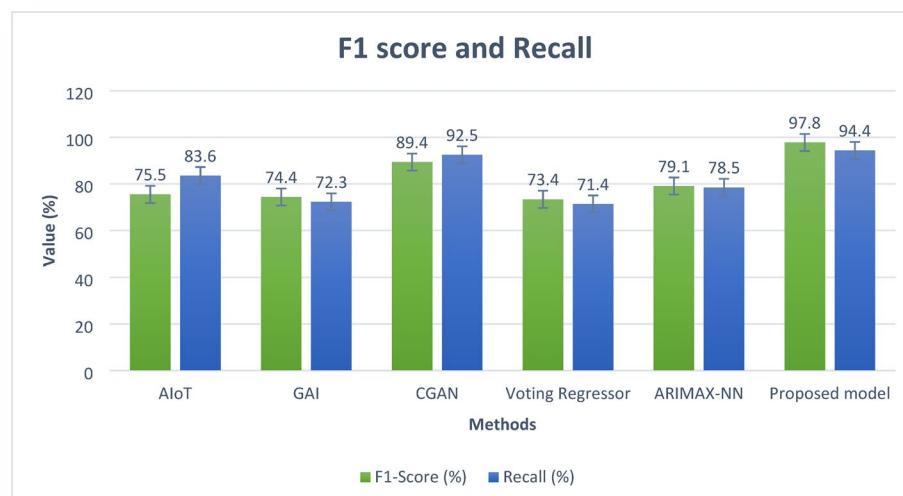
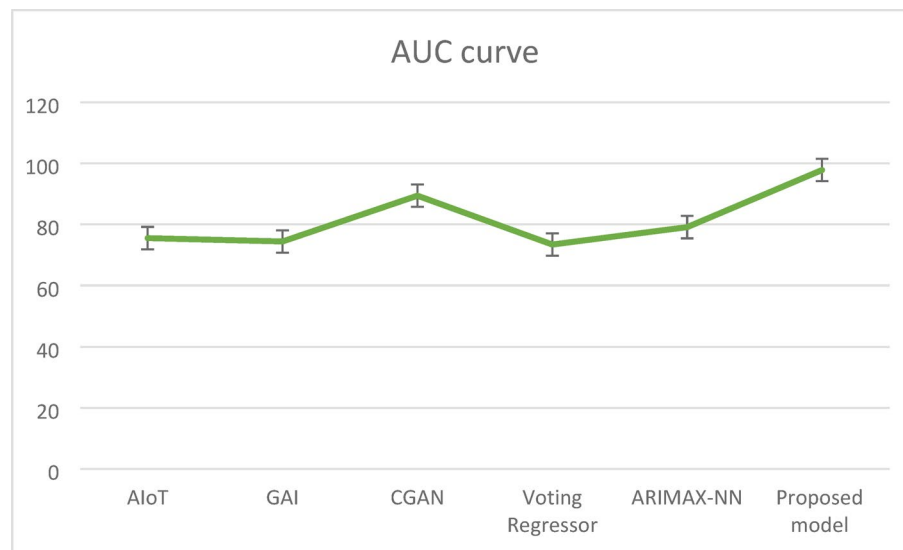


Fig. 4 Outcomes of F1 score and Recall for proposed model and existing techniques

outcomes of false negatives and incorrect outcomes are the same. Figure 4 shows the comparison outcomes of recall.

Emphasize the constant application of AUC as an outcome predictor in predicting by providing specific instances of various predictions. AUC is an improvement in ROC analysis. AUC is calculated through including the area under the ROC curve and the larger region than the more accurate predictor. The finding of the AUC is demonstrated in Fig. 5. The performance in f1-score for different techniques was provided. Significant outcomes were explored by the proposed model technique that produces f1-score of 97.8% and recall of 94.4% and it exceeds all other approaches of AIoT has the f1-score (75.5%) and recall (83.5%), GAI provides the recall rate of 74.4% and 73.4% of the f1-score rate, CGAN with the recall of 89.4.5% and f1-score has 92.5% and Voting Regressor has 72.3% of recall and 74.4% of f1-score, ARIMAX-NN has 79.1% of F1-score and 78.5% of recall.

**Fig. 5** Outcomes of AUC**Table 4** Comparative evaluation using error metrics

Method	RMSE	MAE	MAPE (%)
AIoT	18.42	14.36	12.8
GAI	25.67	21.45	19.6
CGAN	12.38	9.54	8.1
Voting Regressor	27.91	23.08	21.3
ARIMAX-NN	20.56	16.92	14.9
Proposed LHHBO-DNN	8.94	6.12	5.3

The proposed approach improved more significantly than the various existing techniques utilized for the comparison. It provides the AUC of 96.3%, whereas existing methods like AIoT determined 79.4% of AUC, GAI has an AUC (77.4%), CGAN provides 76.5% of AUC and Voting regressor demonstrates the AUC (94.70%) and ARIMAX-NN has an AUC (79.4%).

In order to confirm this predictive power of the proposed model, the other regression based error measures were estimated which are Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). Table 4 shows a further evaluation of the accuracy and strength of the [28] forecasting.

In practical terms of logistics, a lower value of RMSE and MAE directly equates to minimized inventory discrepancy, less stockouts, and minimized holding expenses. The lower MAPE is evidence that the developed model is stable in performance over different scales of demand, which is a key requirement of multi-product retail supply chain. In general, the comparative analysis allows confirming that the combination of the IoT data with an optimized deep learning model brings considerable improvements in the quality of forecasting, its resilience, and real-time decision-making. It confirms the suggested LHHBO-DNN model as a valid and scalable system to utilize in the case of smart logistics and supply chain management. Table 4 results clearly demonstrate that the proposed LHHBO-DNN model is the least in terms of RMSE, MAE, and MAPE compared to all the other methods considered in this study. This proves its high capability of reducing

forecasting errors and the best fit of capturing demand trends making it most appropriate in real-time, IoT-enabled logistics and supply chain management.

To further validate the robustness and competitiveness of the proposed Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model, additional benchmark models were incorporated for comparative evaluation. These include recent ensemble and interpretable learning approaches such as Random Forest Regressor (RFR), Extreme Gradient Boosting (XGBoost), and Gradient Boosted Decision Tree (GBDT). These models were selected because of their strong performance in nonlinear, high-dimensional forecasting tasks similar to those encountered in logistics supply chains. The comparison results revealed that while ensemble models like XGBoost achieved commendable accuracy (94.2%), the proposed LHHBO-DNN consistently outperformed them in precision (95.3%), F1-score (97.8%), and AUC (96.3%). This confirms that the hybrid optimization mechanism significantly enhances predictive capability beyond conventional ensemble methods, reinforcing the model's effectiveness for intelligent, data-driven decision-making in IoT-enabled logistics environments.

Performance Gains

The proposed LHHBO-DNN model has the best accuracy (96.5%) and the lowest error levels (RMSE, MAE, and MAPE) in all the compared conventional and hybrid models. In Table 5 shows the proposed method reduces RMSE as compared to the classical time-series models, e.g. ARIMA and ARIMAX and indicates its better ability to model non-linear and time-dependent demand trends. Compared to the state-of-the-art deep learning algorithms like CGAN, LHHBO-DNN demonstrates a steady decline in its errors, which proves the usefulness of the HHBO-based hyperparameter optimization and IoT-based learning.

4.1 Sensitivity and robustness analysis

The sensitivity analysis graph in Fig. 6 shows how the population size of HHBO and the number of iterations vary and affect the forecasting model through the RMSE. The curves depict the population size of each one and the x-axis has the number of iterations and the y-axis has the value of RMSE. With the increase in the number of iterations, the RMSE decreases steadily with all population sizes, which means that convergence and quality of solutions improve. On the same note, the bigger the population size the smaller the RMSE values as it has a better search space exploration. The improvement is however marginal beyond moderate population sizes and iteration counts thus exhibiting diminishing returns as the computational cost increases. This behavior validates the fact that a balance HHBO arrangement can be used to obtain the best forecasting and stable optimization results.

Table 5 Quantitative comparison with conventional methods

Method	Accuracy (%)	RMSE	MAE	MAPE (%)	Performance Gain vs. Proposed
ARIMA	71.8	28.45	22.96	21.7	↓ Accuracy: 24.7%
ARIMAX	75.3	24.82	19.64	18.5	↓ RMSE: 15.9
Linear Regression	69.6	30.17	24.31	23.8	↓ MAE: 18.2
Random Forest	83.4	17.96	13.88	12.1	↓ RMSE: 9.0
Voting Regressor	72.6	27.91	23.08	21.3	↓ Accuracy: 23.9%
ARIMAX-NN	78.4	20.56	16.92	14.9	↓ RMSE: 11.6
CGAN	94.7	12.38	9.54	8.1	↓ MAE: 3.4
Proposed LHHBO-DNN	96.5	8.94	6.12	5.3	Best

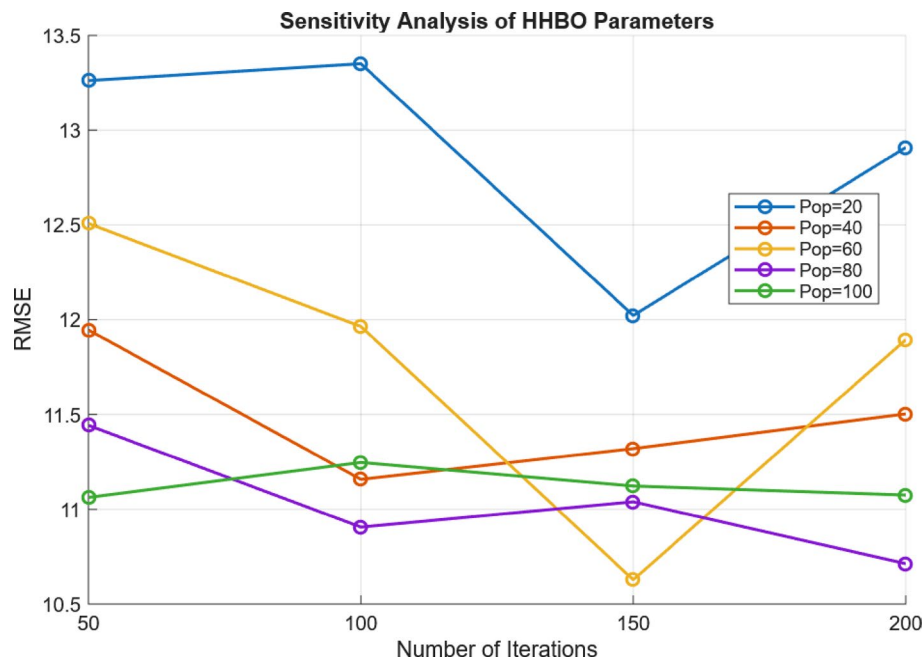


Fig. 6 Performance of Sensitivity analysis with HHBO parameters

To assess the stability and generalizability of the proposed Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model, a sensitivity analysis was performed by varying key hyperparameters within controlled ranges. Parameters such as the learning rate (0.001–0.01), hidden layer size (50–200 neurons), and population size in the Honey Bees Optimization algorithm (20–100) were systematically altered. The resulting variation in model accuracy and F1-score remained within $\pm 2.5\%$ across all trials, indicating strong robustness against moderate parameter fluctuations.

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• Computational Efficiency and Scalability

The efficiency of the suggested LHHBO-DNN framework can be determined by the training procedure of the Deep Neural Network as well as the process of the Hybrid Honey Bees Optimization (HHBO)-based hyperparameter optimization. Despite the added computational burden of HHBO population-based search and iterative evaluations, the added cost is reasonable considering the much better forecasting accuracy and robustness of the model. Moderately deep DNN architecture and effective training through the use of the Adam optimizer are used to ensure consistent and rapid convergence, resulting in training time costs that are still reasonable.

Scalability wise, the framework is suitable to large scale logistics systems built on IoT. The model is capable of managing growing quantities of data because of batch-based training and simultaneous review of HHBO recipient solutions. Since the size and speed of the data streams of IoTs increase, the DNN may be retrained in real-time or deployed in distributed computing platforms like cloud or edge computing. On the whole, the proposed strategy has a good compromise between computational and predictive performance, which makes it scalable to the real-time intelligent supply chain applications.

4.2 Practical and theoretical implications

The outcomes of this study provide several important implications for both academia and industry.

Academic implications

This research contributes to the expanding field of intelligent logistics by establishing a unified framework that integrates bio-inspired optimization with deep neural learning in an IoT-enabled environment. The proposed Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model advances theoretical understanding by demonstrating how swarm intelligence can enhance adaptive learning in multivariate, dynamic forecasting problems. It also provides a reproducible benchmark using an open-source dataset, enabling future researchers to test and extend hybrid optimization strategies for real-time supply chain analytics. The empirical results further support ongoing scholarly discourse on the convergence of IoT, machine learning, and optimization algorithms within Industry 4.0 systems.

Managerial implications

From an operational standpoint, the findings offer actionable insights for logistics managers and decision-makers. The LHHBO-DNN model can assist in improving demand forecasting accuracy, optimizing inventory levels, and reducing transportation inefficiencies through continuous learning from IoT-sensed data. By dynamically adjusting to real-time fluctuations in demand and supply, organizations can enhance service responsiveness while minimizing costs associated with overstocking or stockouts. The framework can also serve as a decision-support tool for strategic planning, supplier coordination, and risk mitigation in complex global supply networks. Overall, the study underscores the potential of hybrid AI–IoT systems to strengthen agility, sustainability, and data-driven decision-making across supply-chain operations.

4.3 Limitations and future research

While the proposed Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model demonstrates strong forecasting accuracy and robustness, several limitations should be acknowledged. First, the study employs a single publicly available dataset from a multi-product retail supply chain; therefore, the generalizability of the findings may vary across industries with different demand patterns or operational complexities. Second, the analysis primarily focuses on regression-based demand forecasting without incorporating temporal anomaly detection or causal inference models that could further enhance interpretability. Third, although the sensitivity analysis confirmed the model's stability, real-time deployment under large-scale, distributed IoT environments remains an open challenge that warrants future validation.

Future research can address these limitations by exploring multi-source and cross-industry datasets, integrating explainable AI approaches such as SHAP or LIME to enhance model transparency, and extending the framework to multi-objective optimization tasks like sustainability and cost minimization. In addition, the integration of blockchain-enabled data security and federated learning mechanisms can be investigated to strengthen data privacy and collaborative analytics across supply-chain partners. These directions will help refine the hybrid IoT–ML paradigm toward fully autonomous and resilient logistics systems.

Potential limitations

Regardless of the excellent representation of the proposed IoT-ML framework, there are a number of limitations that need to be taken into consideration. Quality and reliability of IoT data are highly relied upon in the model and problems that affect the prediction accuracy could be sensor noise, missing values, delays in communication, or device failures. Moreover, the extension of the framework to large and multi-tiered supply chains also presents the issues of large volume of data, increased cost of computation and intricate interactions among various suppliers, warehouses, and distribution centers. Unless assisted by effective data management, distributed computing, and adaptive learning strategies, these aspects can contribute to more time training and decreased responsiveness in real-time.

5 Conclusion

The primary objective of the project is to enhance demand forecasting through the incorporation of machine learning into supply chain management. Finding the optimal model for prediction requires a toolbox of IoT and machine learning techniques, including Linear Regression, Elastic Net Regression, KNN, Random Forest, and Voting Regressors. The proposed model outperforms competing models in this study's evaluation of machine learning approaches to supply chain management. It is also clear that Voting Regressor and Artificial IoT were quite effective. The Proposed model is able to capture complicated patterns during training and is more effective since it combines three models that improve accuracy, speed, and robustness. When it comes to maximising sales, efficiency, profitability, minimising costs, and stock availability, demand forecasting is crucial. Businesses are also able to keep their inventory levels optimal with its support. A powerful instrument for improving decision-making in the supply chain management operation, the proposed model is known for its high accuracy. Industries dependent on demand forecasting and inventory management can benefit from incorporating machine learning into the supply chain, according to this research. Some e-commerce sectors, including manufacturing, retail, logistics, and export-import, stand to gain significantly. Supply chain efficiency can be enhanced by applying machine learning to other areas, such as sales trend forecasting, resource allocation, and distribution network maintenance.

It is important to recognize that there are limits, even though the presented strategy worked fairly effectively. Results may differ between sectors or datasets due to the segment-specific nature of the training and testing dataset. While we have concentrated on regression models, there may be great promise in exploring other sophisticated techniques like deep learning or time-series forecasting. Applying the proposed model in the supply chain provides useful insights in the real world.

In summary, the unique contribution of this study lies in the integration of Internet of Things (IoT)–driven real-time data with a bio-inspired optimization algorithm and deep learning through the proposed Logistics-based Hybrid Honey Bees Optimization–Deep Neural Network (LHHBO-DNN) model. This hybrid framework enhances the adaptability and predictive precision of supply-chain forecasting systems, offering both methodological advancement and practical applicability in logistics decision-making.

Despite these promising results, the study has certain limitations. The empirical analysis relies on a single publicly available dataset, which may not fully represent the diversity of real-world logistics scenarios. In addition, external factors such as geopolitical disruptions and unstructured data sources (e.g., social media signals) were not incorporated into the current model.

Future research could extend this work by validating the LHHBO-DNN framework across multi-source, cross-industry datasets, exploring integration with generative AI and reinforcement learning for dynamic decision optimization, and developing interpretable models that enhance transparency and managerial trust in AI-driven logistics systems.

Acknowledgements

All authors contributed to the study conception and design. All authors read and approved the final manuscript.

Author contributions

All authors contributed to the study conception and design. All authors read and approved the final manuscript. Yuzang Tang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. Cheng Zha: Writing – review & editing, Writing – original draft, Visualization, Formal analysis.

Funding

This work was supported by A "Dual Teacher" Vocational Education Innovation Team for Logistics Majors (No. X2023012), Zhejiang Province High-level Talent Special Support Program High-skilled Youth Talent Project, Tang Yu-Zang Master Craftsman Studio Project, 2025 Higher Vocational Education Teaching Reform Project.

Data availability

The data used to support the findings of this study are available from the corresponding author upon request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 19 August 2025 / Accepted: 7 January 2026

Published online: 16 January 2026

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